GENERALIZED LELONG NUMBERS DETERMINE VALUATIVE TRANSFORMS

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Abstract. Boucksom, Favre, and Jonsson proved that equality of the valuative transforms of two plurisubharmonic germs implies equality of all generalized Lelong numbers with tame reference weights. We prove the converse. It is enough to test against the analytic weights defined by primary ideals. The proof first polarizes the resulting diagonal intersection identities and then applies a strict relative Hodge index statement on each birational model.

1. Introduction

Let o denote the origin of $\mathbb{C}^n$, and let $u, v : (\mathbb{C}^n, o) \to \mathbb{R} \cup \{-\infty\}$ be plurisubharmonic (psh) germs. If φ is a psh weight with an isolated pole at o, Demailly's generalized Lelong number is

$$\nu_\varphi(u) = \big(dd^c u \wedge (dd^c\varphi)^{n-1}\big)(\{o\}). \tag{1}$$

The definition, comparison theorem, and invariance under bounded changes of the weight are established in [Dem87].

We use the terminology of [BFJ08, Section 5.3]. A *psh weight* is a psh germ φ with an isolated singularity at o such that e^φ is continuous. It is *tame* if there is a constant $C > 0$ such that, for every $t > 0$ and every $f \in \mathcal{J}(t\varphi)$,

$$\log|f| \le (t - C)\varphi + O(1). \tag{2}$$

Because φ is negative near its pole, inequality (2) says that the singularities detected by the multiplier ideals approximate φ with an error uniformly bounded in t. We shall not use the definition directly; it is included to make clear which class of test weights occurs in the theorem.

Boucksom–Favre–Jonsson associate to u a valuative transform $\widehat{u}$. The same information can be written as a nef Weil b-divisor $Z(u)$ over o: the coefficient of $Z(u)$ along an exceptional prime divisor is the negative of the corresponding generic Lelong number. We recall the construction in

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section 2. The language of b-divisors goes back to Shokurov; the local form used here is developed in [BFJ08, Sections 1–4]; see also [Sho03].

The following four conditions appear in [BFJ08, Theorem A]:

(1) u and v have the same Lelong number at every infinitely near point;
(2) $\mathcal{J}_+(tu) = \mathcal{J}_+(tv)$ for every $t > 0$;
(3) the relative types of u and v with respect to every tame maximal psh weight agree;
(4) $\nu_\varphi(u) = \nu_\varphi(v)$ for every tame psh weight φ.

Here $\mathcal{J}_+(w)$ is the stationary value of $\mathcal{J}((1+\varepsilon)w)$ as $\varepsilon \downarrow 0$. Boucksom–Favre–Jonsson proved that (1), (2), and (3) are equivalent and imply (4). They left (4)⇒(1) open and proposed the construction of fundamental solutions for the formal Monge–Ampère operator as one possible approach; see [BFJ08, p. 455 and p. 457].

We prove this remaining implication. The statement below is slightly stronger: only weights with analytic singularities coming from primary ideals are needed.

Theorem 1.1. *Let $u, v : (\mathbb{C}^n, o) \to \mathbb{R} \cup \{-\infty\}$ be psh germs. The following conditions are equivalent.*

(i) *u and v have the same Lelong number at every infinitely near point.*
(ii) *$\nu_\varphi(u) = \nu_\varphi(v)$ for every tame psh weight φ.*
(iii) *For every $\mathfrak{m}_o$-primary ideal $\mathfrak{a} = (f_1, \ldots, f_N) \subset \mathcal{O}_{\mathbb{C}^n,o}$,*

$$\nu_{\varphi_\mathfrak{a}}(u) = \nu_{\varphi_\mathfrak{a}}(v), \qquad \varphi_\mathfrak{a} := \frac{1}{2}\log\left(\sum_{j=1}^{N} |f_j|^2\right).$$

Consequently, all four conditions of [BFJ08, Theorem A] *are equivalent.*

Here is the proof in outline. Put $W = Z(u) - Z(v)$. For a primary ideal $\mathfrak{a}$, the analytic weight $\varphi_\mathfrak{a}$ satisfies

$$\nu_{\varphi_\mathfrak{a}}(u) = -\left\langle Z(u), Z(\mathfrak{a})^{n-1}\right\rangle. \tag{3}$$

Thus condition (iii) gives diagonal equations

$$\left\langle W, Z(\mathfrak{a})^{n-1}\right\rangle = 0. \tag{4}$$

Products of ideals turn sums of their divisors into further test divisors. Expanding (4) for these products recovers every mixed intersection. The ideal divisors span the Cartier b-divisors. On a fixed model X_π, we may therefore test W against its own incarnation W_π and against a relatively ample exceptional class H. This gives

$$W_\pi^2 \cdot H^{n-2} = 0.$$

A strict relative Hodge index theorem forces $W_\pi = 0$. Since the model was arbitrary, $W = 0$.

Each input is separated below. Sections 2 and 3 review the b-divisor and intersection formalism, section 4 proves the exact strict Hodge statement required, and section 5 writes out polarization before the proof of the theorem.

2. Birational models and valuative divisors

2.1. Models above the origin. We work with a cofinal system of smooth projective modifications

$$\pi : X_\pi \longrightarrow (\mathbb{C}^n, o)$$

that are isomorphisms away from o. Replacing a modification by a smooth dominating one does not lose any divisorial valuation. Resolution of singularities and principalization provide such dominating models. The local Riemann–Zariski space is constructed in [BFJ08, Sections 1.1–1.2].

Write $\mathrm{Exc}(\pi)$ for the exceptional locus and set

$$\mathrm{Div}(\pi) := \bigoplus_{E \subset \mathrm{Exc}(\pi)} \mathbb{R}E. \tag{5}$$

Every exceptional prime is proper over o. Since X_π is smooth, an exceptional divisor is determined by its relative numerical class, and

$$\mathrm{Div}(\pi) \simeq N^1(X_\pi/\mathbb{C}^n)_\mathbb{R}; \tag{6}$$

see [BFJ08, Section 1.2]. If $\mu : X_{\pi'} \to X_\pi$ is a higher model, exceptional divisors admit pushforward and pullback, with $\mu_*\mu^*D = D$.

Definition 2.1. A *Weil b-divisor* Z over o is a compatible collection $Z = (Z_\pi)_\pi$ with $Z_\pi \in \mathrm{Div}(\pi)$ and $\mu_* Z_{\pi'} = Z_\pi$ whenever π' dominates π. The divisor Z_π is the *incarnation* or *trace* of Z on X_π.

A *Cartier b-divisor* C is a Weil b-divisor for which some model X_π satisfies $C_{\pi'} = \mu^* C_\pi$ on every higher model. Such a model is a *determination* of C.

Thus $D \in \mathrm{Div}(\pi)$ defines a Cartier b-divisor by pullback to higher models; denote it by $\overline{D}$ when needed. A Weil b-divisor need not be determined on any finite model. These are the projective- and inductive-limit definitions in [BFJ08, Section 1.2]. Two Weil b-divisors are equal precisely when their incarnations agree on every model.

For later use, we also recall positivity. A Cartier b-divisor C determined on X_π is *nef over o* if

$$C_\pi \cdot \Gamma \geq 0 \tag{7}$$

for every complete curve $\Gamma \subset \pi^{-1}(o)$. This criterion is [BFJ08, Proposition 2.2]. A nef Weil b-divisor is, by definition, a coefficientwise limit of nef Cartier b-divisors. Thus "nef" here is a relative condition over the germ, not nefness on a compact ambient variety.

2.2. **The divisor attached to a psh germ.** Let E be an exceptional prime on X_π. At a very general point of E, $u \circ \pi$ has a generic Lelong number $\nu_E(u)$. Define

$$\operatorname{ord}_E Z(u) := -\nu_E(u). \tag{8}$$

Equivalently, if

$$dd^c(u \circ \pi) = \sum_E \nu_E(u)[E] + R_\pi$$

is the Siu decomposition and R_π does not charge an exceptional prime, then $[Z(u)_\pi] = -\sum_E \nu_E(u)[E]$. This is the construction in [BFJ08, Section 5.2].

Proposition 2.2 (Boucksom–Favre–Jonsson)**.** *For every psh germ u, the Weil b-divisor $Z(u)$ is nef.*

Proof. This is [BFJ08, Theorem 5.1]. The approximation behind it is useful to recall. The ideal divisors $k^{-1}Z(\mathcal{J}(ku))$ are nef Cartier b-divisors and converge coefficientwise to $Z(u)$. Formula (5.3) of [BFJ08] bounds $Z(u)$ between the divisor of its multiplier ideal and that divisor minus the log-discrepancy function. Division by k makes the error tend to zero. Nef Weil b-divisors are defined as the closure of the nef Cartier cone; see [BFJ08, Sections 2.1 and 3.4]. □

Remark 2.3 (Signs). The coefficients in (8) are nonpositive. This is the convention of [BFJ08]. For example, on the blowup of o, $-E$ is relatively ample and has positive degree on curves contracted by the blowup.

2.3. **Infinitely near points.** Let $p \in \pi^{-1}(o)$ and let $\mu : \mathrm{Bl}_p X_\pi \to X_\pi$ be the blowup of the smooth point p, with exceptional divisor F_p. In local coordinates centered at p, the order along F_p is the ordinary multiplicity at p. Its psh counterpart is

$$\nu_{F_p}(u) = \nu^L(u \circ \pi, p), \tag{9}$$

where ν^L is the usual Lelong number. This follows directly from the definition after pullback by the ordinary blowup and is the observation used in [BFJ08, Section 5.6]. Consequently,

$$Z(u) = Z(v) \Longleftrightarrow \nu^L(u \circ \pi, p) = \nu^L(v \circ \pi, p) \text{ for all } (\pi, p). \tag{10}$$

For the forward direction, compare coefficients on the blowup at p. For the reverse direction, test at very general points of every exceptional prime and use (8).

3. Primary ideals, analytic weights, and intersections

3.1. **The Cartier b-divisor of an ideal.** Let $\mathfrak{a} \subset \mathcal{O}_{\mathbb{C}^n,o}$ be $\mathfrak{m}_o$-primary. For each exceptional prime E, set

$$\operatorname{ord}_E Z(\mathfrak{a}) := -\operatorname{ord}_E(\mathfrak{a}), \qquad \operatorname{ord}_E(\mathfrak{a}) := \min_{f \in \mathfrak{a}} \operatorname{ord}_E(f). \tag{11}$$

On a log resolution π of $\mathfrak{a}$,

$$\mathfrak{a}\mathcal{O}_{X_\pi} = \mathcal{O}_{X_\pi}\big(Z(\mathfrak{a})_\pi\big). \tag{12}$$

On higher models this divisor pulls back. Thus $Z(\mathfrak{a})$ is Cartier; see [BFJ08, Section 1.2]. For blowups and their tautological line bundles, see [Har77, Chapter II, Section 7]. The sheaf in (12) is generated relative to the base, so $Z(\mathfrak{a})$ is nef; compare [BFJ08, Proposition 2.2].

Valuation additivity gives, for primary ideals $\mathfrak{a}, \mathfrak{b}$,

$$Z(\mathfrak{a}\mathfrak{b}) = Z(\mathfrak{a}) + Z(\mathfrak{b}), \tag{13}$$

and more generally

$$Z(\mathfrak{a}_1^{k_1}\cdots\mathfrak{a}_r^{k_r}) = \sum_{i=1}^{r} k_i Z(\mathfrak{a}_i). \tag{14}$$

These are equations (2.1) of [BFJ08]. A product of $\mathfrak{m}_o$-primary ideals remains primary because its radical is $\mathfrak{m}_o$.

Proposition 3.1 (Ideal divisors span)**.** *The real vector space of Cartier b-divisors is spanned by the divisors $Z(\mathfrak{a})$ of $\mathfrak{m}_o$-primary ideals. Thus every Cartier b-divisor has an expression*

$$C = \sum_{j=1}^{N} \lambda_j Z(\mathfrak{a}_j), \qquad \lambda_j \in \mathbb{R}. \tag{15}$$

Proof. This is the real form of [BFJ08, Proposition 1.14]. We recall its geometric content. If an integral Cartier b-divisor is determined by an integral exceptional divisor D on X_π, choose an integral relatively ample exceptional divisor A. For $m \gg 1$, both $\mathcal{O}_{X_\pi}(mA)$ and $\mathcal{O}_{X_\pi}(D+mA)$ are generated relative to the base. Thus $D = D_1 - D_2$ with $D_1 = D + mA$ and $D_2 = mA$, and both line bundles $\mathcal{O}_{X_\pi}(D_i)$ are relatively generated. Their pushforwards are primary ideals $\mathfrak{a}_i$, and $D_i = Z(\mathfrak{a}_i)_\pi$. Hence integral Cartier b-divisors are generated as an abelian group by ideal divisors. Tensoring with $\mathbb{R}$ gives (15). ☐

Remark 3.2. The coefficients λ_j may be negative. Thus proposition 3.1 concerns the full vector space, not only the nef cone. This permits the later test against the signed Cartier divisor $\overline{W_\pi}$.

3.2. **The analytic weight of an ideal.** For generators $\mathfrak{a} = (f_1, \ldots, f_N)$, set

$$\varphi_{\mathfrak{a}} = \frac{1}{2}\log\left(\sum_{j=1}^{N} |f_j|^2\right). \tag{16}$$

Two generating sets change this weight by a bounded function: each generator of either set is a bounded holomorphic linear combination of the other set. Demailly's comparison theorem implies that the generalized Lelong number is unchanged; see [Dem87].

Since $\mathfrak{a}$ is primary, its zero set is $\{o\}$. Moreover $e^{\varphi_{\mathfrak{a}}} = (\sum_j |f_j|^2)^{1/2}$ is locally Lipschitz. Lemma 5.10 of [BFJ08] says that a psh germ with Hölder-continuous exponential is tame. Hence $\varphi_{\mathfrak{a}}$ is a tame psh weight. Its generic Lelong number along E is $\mathrm{ord}_E(\mathfrak{a})$, so

$$Z(\varphi_{\mathfrak{a}}) = Z(\mathfrak{a}). \tag{17}$$

3.3. **Intersection products.** For $D_1, \dots, D_n \in \mathrm{Div}(\pi)$, their local intersection number $D_1 \cdots D_n$ is computed on X_π. It is defined because the divisors are supported over o, so the resulting zero-cycle is proper. The product is symmetric, multilinear, and preserved on higher models by the projection formula; see [BFJ08, Section 4.1, Proposition 4.2] and [Ful98, Section 2.3]. Hence for Cartier b-divisors determined on X_π,

$$\langle C_1, \dots, C_n \rangle := C_{1,\pi} \cdots C_{n,\pi}. \tag{18}$$

Boucksom–Favre–Jonsson extend this product to nef Weil b-divisors by monotone approximation; see [BFJ08, Definition 4.3 and Proposition 4.4]. We need only the following finite case.

More explicitly, if $Z_1, \dots, Z_n$ are nef Weil b-divisors, they set

$$\langle Z_1, \dots, Z_n \rangle := \inf_{C_i \geq Z_i} \langle C_1, \dots, C_n \rangle, \tag{19}$$

where the infimum is taken over nef Cartier b-divisors C_i dominating Z_i coefficientwise. Monotonicity of intersection for nef Cartier divisors makes this compatible with (18). The value can in general be $-\infty$, but it is finite in the one-Weil-factor situation used below.

Proposition 3.3 (One Weil factor)**.** *Let Z be a nef Weil b-divisor and $C_2, \dots, C_n$ Cartier b-divisors. Then $\langle Z, C_2, \dots, C_n \rangle$ is finite and multilinear in the Cartier variables. If the C_i are determined on X_π, then*

$$\langle Z, C_2, \dots, C_n \rangle = Z_\pi \cdot C_{2,\pi} \cdots C_{n,\pi}. \tag{20}$$

The same holds when Z is a difference of two nef Weil b-divisors.

Proof. Finiteness with at most one non-Cartier nef factor is stated after [BFJ08, Proposition 4.9]; its proof also gives (20). By proposition 3.1, a Cartier factor is a signed linear combination of nef ideal divisors, so additivity extends the product to arbitrary Cartier factors. For $Z = Z^+ - Z^-$, define the pairing as the difference of the two finite pairings with Z^+ and Z^-. The asserted properties follow term by term. □

Remark 3.4 (Mixed multiplicity). For primary ideals,

$$-\langle Z(\mathfrak{a}_1), \dots, Z(\mathfrak{a}_n) \rangle = e(\mathfrak{a}_1, \dots, \mathfrak{a}_n). \tag{21}$$

This is formula (4.3) of [BFJ08], based on [Ram74, Ree84]. It is not logically needed below, but explains why ideal multiplication supplies the polarization data.

3.4. **Generalized Lelong numbers as intersections.**

Proposition 3.5. *For every psh germ u and every $\mathfrak{m}_o$-primary ideal $\mathfrak{a}$,*

$$\nu_{\varphi_{\mathfrak{a}}}(u) = -\left\langle Z(u), Z(\mathfrak{a})^{n-1} \right\rangle . \tag{22}$$

Proof. Theorem B of [BFJ08] gives, for every tame weight φ,

$$\nu_\varphi(u) = \int_{\mathcal{V}} -\widehat{u}\ \mathrm{MA}(\widehat{\varphi}), \tag{23}$$

where $\mathcal{V}$ is the normalized valuation space. Proposition 4.6 of [BFJ08] defines the formal Monge–Ampère measure by

$$\int_{\mathcal{V}} \widehat{u}\ \mathrm{MA}(\widehat{\varphi}) = \left\langle Z(u), Z(\varphi)^{n-1} \right\rangle . \tag{24}$$

The measure is well-defined here because $Z(\varphi_{\mathfrak{a}})$ is Cartier; finiteness also follows from proposition 3.3. Substitute $\varphi = \varphi_{\mathfrak{a}}$ into (23), then use (24) and (17). This gives (22). The minus sign agrees with the convention in remark 2.3. □

Remark 3.6. This proposition contains the deep analytic input: Demailly approximation and the comparison between analytic and formal Monge–Ampère masses in [BFJ08]. After (22), the proof is finite-dimensional intersection theory and an elementary polynomial argument.

4. Strict relative Hodge separation

The Hodge input must be strict. A semidefinite statement would not show that a divisor with square zero vanishes.

Lemma 4.1 (Strict relative Hodge index)**.** *Let $\pi : X_\pi \to (\mathbb{C}^n, o)$ be a smooth projective modification, $n \geq 2$. Let $H \in \mathrm{Div}(\pi)_{\mathbb{Q}}$ have a π-ample relative numerical class. Then*

$$Q_H(D_1, D_2) := D_1 \cdot D_2 \cdot H^{n-2} \tag{25}$$

is negative definite on $\mathrm{Div}(\pi)$*. Thus*

$$D \neq 0 \Longrightarrow D^2 \cdot H^{n-2} < 0. \tag{26}$$

Proof. We divide the proof into four steps.

Step 1: the surface case. If $n = 2$, the intersection matrix of the exceptional curves of a proper birational morphism from a smooth surface to a normal surface germ is negative definite. This is Mumford's negativity theorem [Mum61]. Hence $D^2 < 0$ for every nonzero exceptional real divisor D.

Step 2: an ample surface slice. Assume $n \geq 3$. The ample cone is open in $N^1(X_\pi/\mathbb{C}^n)_{\mathbb{R}}$. By (6), it contains a rational exceptional class. Multiplying H by a positive integer, we may suppose that H is Cartier and relatively very ample. This changes Q_H only by a positive scalar. Relative ampleness is reviewed in [Laz04, Chapter I, Section 1.7].

After shrinking the base, take $n-2$ general relative hyperplane sections $L_1, \ldots, L_{n-2} \in |H|$ and put $S = L_1 \cap \cdots \cap L_{n-2}$. Bertini's theorem makes S smooth near $\pi^{-1}(o)$; see [Har77, Chapter III, Corollary 10.9]. The restriction $\pi|_S$ is proper and birational onto a surface germ. If necessary, factor it through the normalization of that germ. This is the slicing reduction used in [BdFF12, Lemma A.2].

Step 3: restriction is injective. Let E be an exceptional prime. Since $H|_E$ is ample and $\dim E = n-1$,

$$E|_S = E \cdot H^{n-2} \tag{27}$$

is a nonzero curve contracted by $\pi|_S$. If $E \neq F$, then $E \cap F$ has codimension at least two in X_π. Generality of the L_i implies that $E|_S$ and $F|_S$ share no curve component: the expected dimension of $E \cap F \cap S$ is zero.

Write $D = \sum_E d_E E$ and choose E_0 with $d_{E_0} \neq 0$. A curve component of $E_0|_S$ is not a component of any $F|_S$ with $F \neq E_0$, so it has nonzero coefficient in $D|_S$. Therefore

$$D \neq 0 \Longrightarrow D|_S \neq 0. \tag{28}$$

Step 4: apply surface negativity. The components of $D|_S$ are exceptional curves. By Step 1 and (28), $(D|_S)^2 < 0$ when $D \neq 0$. Associativity of intersection gives

$$(D|_S)^2 = D^2 \cdot H^{n-2}.$$

This proves (26) and the lemma. □

Remark 4.2 (Nef versus ample). Lemma A.2 of [BdFF12] gives negative semidefiniteness for merely nef remaining factors. Semidefiniteness permits nonzero null vectors. Relative ampleness ensures that every exceptional prime cuts a nonzero curve and makes (28) available, yielding strict negativity.

Remark 4.3 (Existence of H). Projectivity of π gives a nonempty open ample cone in $N^1(X_\pi/\mathbb{C}^n)_\mathbb{R}$. The isomorphism (6) represents a rational point of this cone by an exceptional $\mathbb{Q}$-divisor. Thus such an H always exists.

5. Polarization of diagonal identities

Lemma 5.1 (Polynomial vanishing on a positive lattice)**.** *If a polynomial P in r real variables vanishes at every point of $\mathbb{N}^r_{>0}$, then $P = 0$.*

Proof. Induct on r. For $r = 1$, a nonzero polynomial has finitely many roots. Write $P = \sum_{j=0}^{d} P_j(x_1, \ldots, x_{r-1}) x_r^j$. After fixing positive integers $k_1, \ldots, k_{r-1}$, the resulting polynomial in x_r has infinitely many roots, so every coefficient $P_j(k_1, \ldots, k_{r-1})$ is zero. Induction gives $P_j = 0$ for every j. □

Lemma 5.2 (Polarization by products of ideals)**.** *Let W be a difference of nef Weil b-divisors. If*

$$\langle W, Z(\mathfrak{a})^{n-1}\rangle = 0 \tag{29}$$

for every $\mathfrak{m}_o$-primary ideal $\mathfrak{a}$, then

$$\langle W, Z(\mathfrak{a}_1), \dots, Z(\mathfrak{a}_{n-1})\rangle = 0 \tag{30}$$

for all primary ideals $\mathfrak{a}_i$.

Proof. Set $r = n-1$ and $A_i = Z(\mathfrak{a}_i)$. For $\mathbf{k} = (k_1, \dots, k_r) \in \mathbb{N}^r_{>0}$, the ideal $\mathfrak{a}(\mathbf{k}) = \mathfrak{a}_1^{k_1} \cdots \mathfrak{a}_r^{k_r}$ is primary and (14) gives $Z(\mathfrak{a}(\mathbf{k})) = \sum_i k_i A_i$. Hence

$$P(\mathbf{k}) := \langle W, (k_1 A_1 + \cdots + k_r A_r)^r\rangle = 0. \tag{31}$$

By multilinearity,

$$P(\mathbf{k}) = \sum_{\alpha_1+\cdots+\alpha_r=r} \frac{r!}{\alpha_1! \cdots \alpha_r!} k_1^{\alpha_1} \cdots k_r^{\alpha_r} \left\langle W, A_1^{\alpha_1}, \dots, A_r^{\alpha_r}\right\rangle . \tag{32}$$

By lemma 5.1, this polynomial is identically zero. The coefficient of $k_1 \cdots k_r$ is $r!\langle W, A_1, \dots, A_r\rangle$, so that mixed intersection vanishes. □

Example 5.3 (Dimension three)**.** For $n = 3$, put $A = Z(\mathfrak{a})$ and $B = Z(\mathfrak{b})$. The diagonal identities for A, B, and $A + B = Z(\mathfrak{a}\mathfrak{b})$ give

$$0 = \langle W, (A+B)^2\rangle = \langle W, A^2\rangle + 2\langle W, A, B\rangle + \langle W, B^2\rangle.$$

The first and last terms vanish, so $\langle W, A, B\rangle = 0$. The general argument is the degree-$(n-1)$ multinomial version of this calculation.

Proposition 5.4 (Extension to all Cartier tests)**.** *Under the hypothesis of lemma 5.2,*

$$\langle W, C_1, \dots, C_{n-1}\rangle = 0 \tag{33}$$

for arbitrary Cartier b-divisors C_i.

Proof. Use proposition 3.1 to write each C_i as a finite real linear combination of divisors $Z(\mathfrak{a}_{ij})$. Multilinearity expands the left side of (33) into finitely many mixed ideal intersections. Every term vanishes by lemma 5.2. □

6. Proof of the main theorem

Proof of theorem 1.1. We prove the implications separately.

Step 1: (i)⇒(ii). By (10), condition (i) is equivalent to $Z(u) = Z(v)$, hence to $\widehat{u} = \widehat{v}$. Theorem B of [BFJ08] expresses generalized Lelong numbers with tame weights by (23). Equal valuative transforms have equal integrals.

Step 2: (ii)⇒(iii). The weight $\varphi_{\mathfrak{a}}$ is tame by [BFJ08, Lemma 5.10], as explained after (16); hence it may be substituted into condition (ii).

Step 3: obtain diagonal intersection identities. If $n = 1$, then $(dd^c\varphi)^0 = 1$ and $\nu_\varphi(u) = (dd^c u)(\{o\}) = \nu^L(u, o)$. A proper modification of a smooth curve germ is an isomorphism, so the conclusion follows. Assume $n \geq 2$ and set

$$W := Z(u) - Z(v). \tag{34}$$

It is a difference of nef Weil b-divisors by proposition 2.2. Using condition (iii) and proposition 3.5,

$$\begin{aligned} 0 &= \nu_{\varphi_{\mathfrak{a}}}(u) - \nu_{\varphi_{\mathfrak{a}}}(v) \\ &= -\langle Z(u), Z(\mathfrak{a})^{n-1}\rangle + \langle Z(v), Z(\mathfrak{a})^{n-1}\rangle \\ &= -\langle W, Z(\mathfrak{a})^{n-1}\rangle. \end{aligned}$$

Therefore

$$\langle W, Z(\mathfrak{a})^{n-1}\rangle = 0 \quad \text{for every primary ideal } \mathfrak{a}. \tag{35}$$

Step 4: pass from diagonal to arbitrary Cartier tests. Lemma 5.2 turns (35) into all mixed ideal identities. Proposition 5.4 then gives

$$\langle W, C_1, \ldots, C_{n-1}\rangle = 0 \tag{36}$$

for all Cartier b-divisors C_i.

Step 5: separate W on a fixed model. Fix an arbitrary smooth projective model X_π. Its trace $W_\pi = Z(u)_\pi - Z(v)_\pi$ is a real exceptional divisor. It defines the Cartier b-divisor $\overline{W_\pi}$ determined on X_π. Choose a relatively ample exceptional rational divisor H as in remark 4.3, also viewed as a Cartier b-divisor.

In (36), take

$$C_1 = \overline{W_\pi}, \qquad C_2 = \cdots = C_{n-1} = H.$$

There are $n-2$ copies of H; for $n = 2$ there are none. We obtain

$$\langle W, \overline{W_\pi}, H^{n-2}\rangle = 0.$$

All Cartier factors are determined on X_π, so the projection formula (20) gives

$$W_\pi^2 \cdot H^{n-2} = 0. \tag{37}$$

By lemma 4.1, a nonzero exceptional divisor has strictly negative square against H^{n-2}. Hence $W_\pi = 0$.

Step 6: return to infinitely near points. The model was arbitrary, so every incarnation of W is zero. Thus $W = 0$ and $Z(u) = Z(v)$. Equation (10) gives equality of all infinitely near Lelong numbers, which is condition (i). □

Remark 6.1 (The surface case). When $n = 2$, (35) is already linear in $Z(\mathfrak{a})$. Since ideal divisors span all Cartier b-divisors, it gives $\langle W, C\rangle = 0$ for all Cartier C. Taking $C = \overline{W_\pi}$ and using Mumford negativity gives $W_\pi = 0$. Polarization is needed only in dimensions at least three.

Remark 6.2 (First-variation interpretation). For ideal singularities in every entry, (21) identifies the intersections with mixed multiplicities. The diagonal functional

$$A \longmapsto -\langle Z(u), A^{n-1}\rangle$$

determines all mixed values by polarization. Relative Hodge separation then recovers $Z(u)$. This explains why fundamental solutions of the formal Monge–Ampère operator are unnecessary here: individual valuations need not be isolated by positive measures once signed Cartier tests are available.

Remark 6.3 (Logical dependencies). The non-elementary inputs are:

(1) nefness of $Z(u)$, [BFJ08, Theorem 5.1];
(2) generation of Cartier b-divisors by ideal divisors, [BFJ08, Proposition 1.14];
(3) the intersection product and projection formula, [BFJ08, Section 4, Propositions 4.2, 4.6, and 4.9];
(4) the generalized Lelong formula, [BFJ08, Theorem B];
(5) surface negativity from [Mum61]; and
(6) the higher-dimensional slicing form from [BdFF12, Lemma A.2].

The polynomial argument and the strict separation step are proved here.

References

[BdFF12] S. Boucksom, T. de Fernex, and C. Favre, *The volume of an isolated singularity*, Duke Math. J. **161** (2012), no. 8, 1455–1520.

[BFJ08] S. Boucksom, C. Favre, and M. Jonsson, *Valuations and plurisubharmonic singularities*, Publ. Res. Inst. Math. Sci. **44** (2008), no. 2, 449–494.

[Dem87] J.-P. Demailly, *Nombres de Lelong généralisés, théorèmes d'intégralité et d'analyticité*, Acta Math. **159** (1987), no. 3–4, 153–169.

[Ful98] W. Fulton, *Intersection theory*, second ed., Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 2, Springer-Verlag, Berlin, 1998.

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York–Heidelberg, 1977.

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry I*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 48, Springer-Verlag, Berlin, 2004.

[Mum61] D. Mumford, *The topology of normal singularities of an algebraic surface and a criterion for simplicity*, Publ. Math. Inst. Hautes Ètudes Sci. **9** (1961), 5–22.

[Ram74] C. P. Ramanujam, *On a geometric interpretation of multiplicity*, Invent. Math. **22** (1973/74), 63–67.

[Ree84] D. Rees, *Generalizations of reductions and mixed multiplicities*, J. London Math. Soc. (2) **29** (1984), no. 3, 397–414.

[Sho03] V. V. Shokurov, *Prelimiting flips*, Proc. Steklov Inst. Math. **240** (2003), no. 1, 75–213.

Declaration on the use of artificial intelligence

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